NMR Tuning and Operation

Quadrature Phase Tunes, Auto Tune and the New Run Controls in PyNMR (UNH fork)

UNH Polarized Target Group

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Note on provenance. This document was drafted with AI assistance (Claude), under the direction of and with technical input from E. Long. It has not undergone the review process required for publication and may contain errors, gaps, or outdated statements. It is included here because it is useful for understanding and running the accompanying code, not as a citable or publication-ready reference.

Abstract

This note is the operator's guide to the tuning and quadrature features added to PyNMR in the UNH fork (`unhpoltarg/jlab_pynmr`) during October–November 2026. PyNMR can now record both quadratures of the deuteron NMR signal, absorption (Real, 0°) and dispersion (Imag, 90°), by stepping the phase detector's reference between events, optionally with the opposite-sign settings 180° and 270° that cancel detector offsets. The operator sets each phase tune roughly by eye; **Auto Tune** then moves them to the true angles using only the background Q-curve, in about 50 s for a Real/Imag pair. The note covers starting PyNMR safely, tuning, taking quadrature data, what lands in which file, and what to do when something goes wrong. An appendix explains why the Q-curve alone fixes the absorptive phase: at every frequency the NMR absorption points along the normal to the Q-curve's trace in the complex plane, which is the origin of the empirical form $\theta(f) = \theta_0 + 2\arctan\big((f - f_0)/\gamma\big)$. Everything here has been checked against the built-in simulator. **Auto Tune, the split plots, the per-angle baselines, the dispersive background and the $64 \times N$ sweep controls have not yet been run on the Hall B box.**

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1 What is new

Feature	What it does for the operator	Where
Python 3.12 env	PyNMR on the DAQ computer runs in `pynmr312`; the old 3.11 env crashed mid-run	§2
Box connections	Each event closes its connection to the FPGA box cleanly; the box is told to stop after errors	§2
Data streamer	Each finished 0° event goes to the online fitter; on by default	§2
NMR window	Centre frequency and half-width set from the Run tab, between events	§4.1
$64 \times N$ sweeps	Sweeps per event and the Tune running average are whole 64-sweep chunks	§4.2
Wing points	The wing fit's size as points per side, with the fitted points marked	§4.3
Phase tunes	Separate Real, Imag, 180° and 270° tunes on the Tune tab	§5
Auto Tune	Refines the hand tunes to the true angles from the Q-curve	§5.2
Auto Calibrate Phase	Measures the phase shifter over its whole range	§5.3
Quadrature runs	Events alternate between tunes: pair, abba or cycle4	§6
Quadrature baselines	One selection of a baseline run gives each angle its own baseline	§6.2
Dispersive background	90°/270° records: background fitted jointly with KK of the absorption	§6.3
Quadrature plots	A 2×3 grid: each processing stage split into 0° and 90°	§6.4

Table 1: The new features and where they are described.

All UNH code lives in `unh/`; every change to a JLab file is marked with a `UNH-HOOK` label and can be removed with `python -m unh.strip_unh`. The quadrature code is in `unh/quadrature/`, its settings in `unh/quadrature/quadrature.yaml`, and its design notes in `notes/quadrature_phase_design.md`.

2 Before you start

Use the Python 3.12 environment. On the DAQ computer (Linux/WSL):

```
conda deactivate          # repeat until no environment is active
conda activate pynmr312
python --version          # must say 3.12 or newer
python pynmr_main.py -p Deuteron
```

On the old Python 3.11 environment PyNMR died silently in the middle of runs, most often with the streamer on. The cause was `Fatal Python error: none_dealloc`, raised while pyqtgraph redrew the plots; Python 3.12 cannot raise it. Stacked conda environments can leave the old Python first on the path, hence the `conda deactivate`. The fitter should also run on Python 3.12.

Never kill PyNMR in the middle of a sweep. The deuteron box (192.168.0.72) stops answering, even on UDP, if PyNMR dies while the box is sweeping, and only a power cycle brings it back. Stop runs with the **Run**/**Stop** button and close PyNMR normally. If you close PyNMR while the Run loop is on, it asks first: choose **Abort! I didn't mean to make this appear!** to keep running; **Ignore. I am an expert and I understand the consequences.** closes it mid-sweep. There is no such warning for the Tune loop, **Auto Tune** or **Auto Calibrate Phase**: stop those before closing. PyNMR now closes every connection cleanly and sends the stop-sweep command after errors and at the end of tuning; on the DAQ computer this cut the connection time from 2.0 s to under 0.2 s per event.

The online fitter. The data streamer is on by default (`unh/data_streamer.yaml`, `enable: True`). It sends each finished deuteron event's fit-subtracted line to LabView-NMR-Fitter on `127.0.0.1:8777`. If the fitter isn't running, PyNMR logs one warning per outage and carries on; it never waits for the fitter. In quadrature runs only the 0° records are sent.

Turning quadrature on for a profile. The quadrature controls appear only for profiles enabled in `unh/quadrature/quadrature.yaml`:

```
profiles:
    Deuteron:
        enable: True
```

The simulator profile `TestSimQuad` has them on already, and runs with no hardware: `python pynmr_main.py -p TestSimQuad`.

3 Concepts

Phase-sensitive detection. The Q-meter's phase channel multiplies the tank voltage V by a reference of phase ϕ and keeps $P_\phi = g\,\mathrm{Re}\big(Ve^{-i\phi}\big) + o$, where g is the detector gain and o a DC offset. With the reference on the absorptive direction the output is the absorption χ'' (**Real,** 0°); a quarter turn later it is the dispersion χ' (**Imag,** 90°). At 180° and 270° the signal flips sign but the offset o does not, so the pair differences

$$I = \tfrac{1}{2}\big(P_0 - P_{180}\big), \qquad Q = \tfrac{1}{2}\big(P_{90} - P_{270}\big) \tag{1}$$

are free of detector offsets.

The phase DAC. The reference phase is set by a Mini-Circuits voltage-controlled phase shifter (SCPHS-51+ on the deuteron box) driven by the box's phase DAC. The Tune tab shows the DAC setting as a percentage of full scale, for example "DAC 51.40 %". **This is a control setting, not a fraction of the signal.** The shifter covers about 456° at 32.7 MHz, but its curve is S-shaped (a few degrees per % near 0, about 10° per % in the middle) and it stops responding above about 80 % DAC. The curve also depends strongly on frequency: a calibration at 32.7 MHz is wrong at 22 MHz.

What the Q-curve can and can't tell. As the frequency is swept, the Q-curve traces a curve in the complex plane (a circle, or nearly a straight line for a heavily damped circuit). Appendix A shows that the NMR absorption at each frequency points along the normal to that trace. So the Q-curve alone fixes the absorptive direction, and **Auto Tune** uses that. Two things it can't fix are left to the hand tunes: whether "0" is θ or $\theta+180°$ (both are absorptive), and whether "90" is $+90°$ or $-90°$ (the sign of the rotation isn't observable from projections). Auto Tune keeps whichever is nearer the operator's hand setting.

The limit on 90°. Projections determine the reference direction only up to a 2×2 linear map, so a 90° angle needs one assumption: that the detector gain changes smoothly with the DAC setting. 0° and 180° (antiparallel) don't need it. PyNMR's numbers are good for choosing and monitoring the tunes; a 0.1° phase should come from the offline complex lineshape fit, which uses the Kramers–Kronig relation between χ' and χ''.

4 Setting up the NMR window and sweep counts

4.1 The NMR window

In NMR Settings on the Run tab, **Center Freq (MHz)** and **Half-width ± (kHz)** set the sweep window; **Apply Window** applies them. When PyNMR is idle the change is immediate; during a run it waits for the next event. The window is saved in the session and restored at the next start. Keep the centre on the deuteron Larmor frequency: **Auto Tune** finds the absorptive phase at the centre frequency.

4.2 Sweeps as $64 \times N$

The box delivers sweeps in chunks of 64, and when asked for 32 it returned 64. Sweep counts are therefore whole chunks:

- **Run tab, Sweeps per Event:** $64 \times N$, with N set by the operator (default 10, i.e. 640 sweeps). Press **Unlock**, change N, press **Set**, as before.
- **Tune tab, Sweeps for Running Average:** $64 \times N$, default $N = 1$. Larger N gives a smoother plot but makes it lag the knobs by about 1.8 s per chunk.
- The tune chunk itself (`tune_per_chunk` in `pynmr_config.yaml`) is 64 sweeps, so the Tune loop, **Auto Tune** and **Auto Calibrate Phase** all use 64-sweep chunks.

4.3 The wing fit: points per side

The yellow curve on Baseline Subtracted is JLab's *Polynomial Fit to Wings* (3rd order by default), which is subtracted to give Fit Subtracted. It is fitted to the outer parts of the sweep only. A

box in the top-right corner of the Baseline Subtracted plot, **Wing points per side**, sets how many points it uses on each side; a dashed red curve on top of the yellow one marks exactly those points.

- The default is what the Analysis tab's fit bounds give (`analysis: wings` in `pynmr_config.yaml`): 148 points per side of 512 for the shared default 0.01/0.30/0.70/0.99, and 122 for the TestSim profiles. JLab's skip of the outermost 1 % on each side is kept.
- Changing N writes the Analysis tab's four bounds (they stay the one setting), redraws the last event's fit at once so the effect can be judged by eye, and applies to every event after it. Records already taken keep the fit they were analysed with.
- Choose N so the dashed red part stays clear of the line: too many points and the fit takes signal for background; too few and the polynomial is poorly pinned down.
- In quadrature the box sits on the Baseline Subtracted Real plot. The 90° plot is dashed over the whole sweep, because dispersive records use the joint KK fit (§6.3), not the wings.
- N goes back to the default when PyNMR restarts.

5 Tuning

Figure 1 shows the Tune tab. Below the diode and phase plots, the **Quadrature Phase Tunes** strip holds a selector for the tune being edited (Real, Imag, and 180°/270° when the sequence uses them) with each tune's DAC setting, the buttons, a result line, and a small plot.

5.1 Tuning procedure

1. Tune the tank (diode) as usual, with the Tune loop running (**Run**).
2. Select **Real (0°)** and set the Phase Tune by eye so the Q-curve looks absorptive. Roughly is enough: within about 30°.
3. Select **Imag (90°)** and set its tune by eye, roughly a quarter turn away. For cycle4, also set 180° and 270°, or leave them empty: with a current calibration (§5.3) PyNMR fills them opposite Real and Imag when cycle4 is turned on.
4. Click **Auto Tune** (§5.2). Check the result line and the plot.
5. Leave the selector on **Real (0°)** when done; runs always restore the Real tune at the end.

Starting values. If the saved session has no usable set of tunes (one is missing, at DAC 0 %, or on the Real tune), PyNMR starts from `default_tunes` in `quadrature.yaml`. These are the settings of the 2026-10-27 Hall B cycle4 run at 32.511 MHz: Real 51.40 %, Imag 44.40 %, 180° 30.39 %, 270° 60.31 %. That run's calibration put them at 0°, −79.9°, −180.0° and +100.0°, so Imag and 270° were about 10° short of quadrature; **Auto Tune** corrects that. A usable saved set (the operator's latest) always wins.

5.2 Auto Tune

Auto Tune refines every tune the current sequence uses (Real and Imag, plus 180° and 270° for cycle4) in one click.

1. It pauses the Tune loop if it is running.

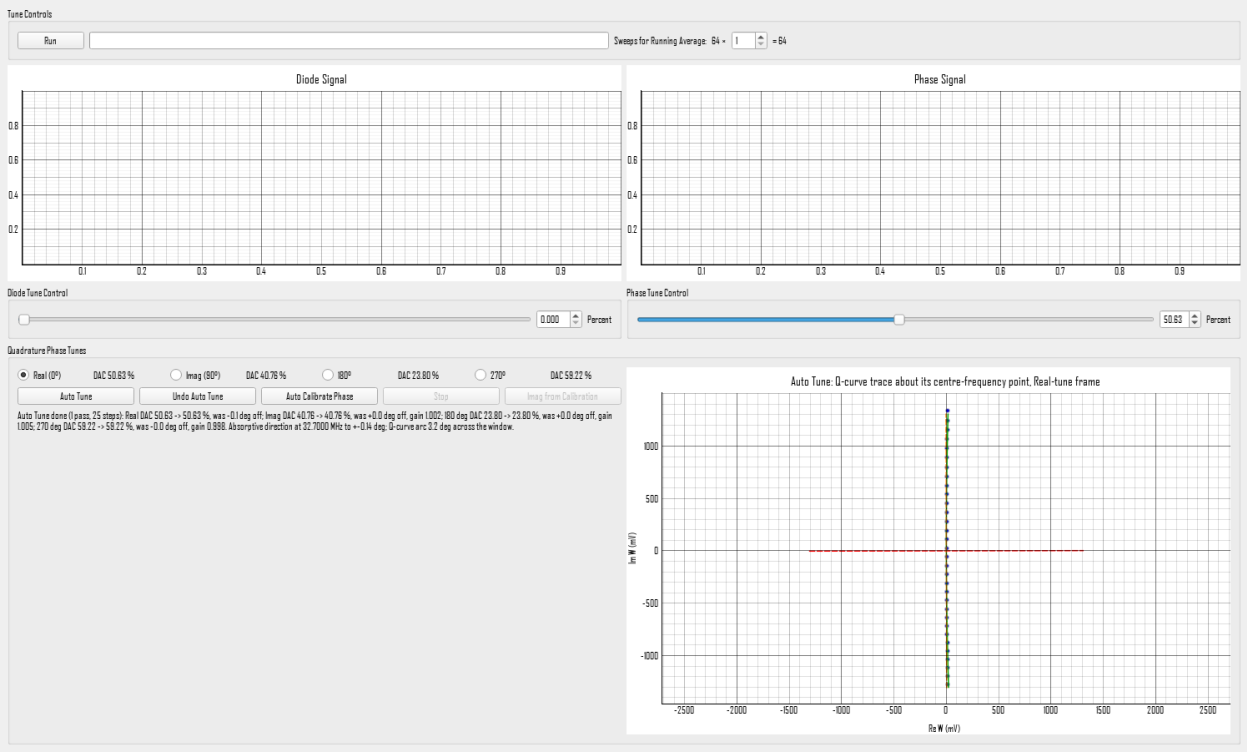


Figure 1: The Tune tab after a cycle4 **Auto Tune** in the simulator. Bottom left: the tune selector with each tune's DAC setting, the **Auto Tune**, **Undo Auto Tune**, **Auto Calibrate Phase**, **Stop** and **Imag from Calibration** buttons, and the result line. Bottom right: the Q-curve trace about its point at the centre frequency, with the fit (green), the 0° absorptive direction (red dashes) and 90° (orange). The simulator's Q-meter is heavily damped, so its trace is nearly a straight line; the Hall B tank traces a visible circle (about 39° of arc in the 2026-10-27 data).

2. For each tune it steps the phase DAC through five settings (-3, -1.5, 0, $+1.5$, $+3$ % of full scale), 64 sweeps each, returning to the Real tune between groups.
3. It fits these steps as a local calibration, which gives the complex Q-curve $W(f)$ and the phase of each setting.
4. It fits the trace of $W(f)$ and finds its normal at the centre frequency: the absorptive direction.
5. It moves each tune to the setting at the true angle. If a target lies outside the five settings around its tune, it repeats, centred on the new values (at most three passes).
6. It then runs one *polish* pass: five tighter settings (-1, -0.5, 0, $+0.5$, $+1$ %) around the new tunes, which measures the new tunes themselves and gives the final answer. If the polish is stopped or a target falls outside its settings, the main pass's answer is kept and the result line says so.
7. It leaves the box on the tune the selector shows and restarts the Tune loop.

A Real/Imag pair takes one pass of 13 steps plus the 13-step polish, about 50 s on the box. A step is one DAC setting visited automatically (64 sweeps); for the operator it is still one click. In the simulator, a cycle4 set with hand tunes 12–32° off took two passes and the polish (75 steps) and landed every tune within 0.15°, which is the statistical error of the absorptive direction itself.

Reading the result line. For example:

> Auto Tune done (1 pass + polish, which moved them by at most 0.31 deg; 26 steps): Real DAC 25.80 → 23.78 %, was +7.6 deg off; Imag DAC 39.30 → 40.76 %, was −10.9 deg off, gain 0.997. Absorptive direction at 32.7000 MHz to ±0.14 deg; Q-curve arc 3.2 deg across the window.

- *was ... deg off*: how far the hand tune was from the true angle.
- *gain*: the detector gain at that tune relative to Real. The shifter's insertion loss varies with the DAC, so values a few per cent from 1 are normal. Large departures mean the 90° angle leans harder on the smooth-gain assumption.
- *to* ±*... deg*: the statistical error of the absorptive direction.
- *arc*: how far the Q-curve turns across the window. It is a few degrees for a heavily damped circuit and tens of degrees for the Hall B tank. Across that arc the absorptive phase itself changes (Appendix A), so no single tune is exactly absorptive at every frequency.

Undo Auto Tune puts back the tunes it started from. The record of each run is saved as `<event_dir>/quad_cal/autotune_<time>.json`; it is never taken for a calibration.

When Auto Tune refuses or fails. It says so in red on the result line and changes nothing.

- *set the ... tune by hand first*, or *the ... tune is at the Real tune*: set each tune roughly to its angle.
- *not converged*: a target was still outside the steps after three passes; set the tunes closer and try again.
- It won't start during a Run; stop the Run first.

With the deuteron line on. Most tuning is done on the bare Q-curve. A deuteron line centred in the window adds an even function along the normal and an odd one along the trace, so to first order it doesn't tilt the normal at the centre. If a strong or off-centre line biases the result, list frequency ranges to leave out in `autotune: mask_khz` (kHz from the centre, for example `[[-60, 60]]`).

5.3 Auto Calibrate Phase

Auto Calibrate Phase measures the shifter over its whole range: 21 DAC settings from 0 to 94 %, up and then down, 64 sweeps each, with returns to the Real tune every 10 steps (about 90 s). It fits the phase and detector gain at every setting relative to Real, and the result line gives each tune's angle and gain, the range, and where the shifter stops responding. Use it:

- at the start of a cooldown, or after any change to the RF hardware;
- after the centre frequency moves by more than 0.05 MHz (the calibration is then marked stale);
- to fill 180°/270° for cycle4, or to set Imag with **Imag from Calibration**.

Calibrations are saved in `<event_dir>/quad_cal/<time>.json` during the scan, so a stop or a crash loses nothing; **Stop** ends it after the current step without fitting. Don't close PyNMR until the line says done.

6 Taking quadrature data

6.1 Starting a quadrature run

1. On the Run tab, choose the sequence next to the **Quadrature** checkbox:
   - **pair** 0/90: the default;
   - **abba** 0/90/90/0: cancels a linear drift between the two quadratures;
   - **cycle4** 0/90/180/270: also cancels the detector offsets, Eq. (1).
2. Tick **Quadrature**. If PyNMR refuses (a tune missing, or two tunes at the same DAC setting), the reason appears in red next to the checkbox.
3. Set Sweeps per Event ($64 \times N$) and press **Run**. Events now alternate through the sequence.
4. Stop by hand with **Stop**. PyNMR puts the Config and the box back on the Real tune.

The checkbox starts unticked at every launch. Each event is still one record in the event file, tagged with its angle. The DAC is written for the next event at the end of the previous one, while the box is idle (a write on the run connection just before the sweeps made 640-sweep events time out).

6.2 Baselines

Each angle needs its own baseline: the Q-curve looks different at each reference phase, and so do the detector offsets. Selecting them is as quick as in normal mode, because one selection of a baseline run is split by angle automatically.

Taking a baseline.

1. Finish tuning first (**Auto Tune**). A baseline belongs to the tunes it was taken at.
2. Move the field so the line is out of the window, as for any baseline (same centre and half-width).
3. On the Run tab, choose the sequence you will take data with, tick **Quadrature**, and set the Event Label to Baseline if you like.
4. **Run** for a few complete cycles (for example 4 cycles of 64×10 sweeps), then **Stop**.

Selecting it.

1. On the Baseline tab, click **Select Current Eventfile** (or **Show Recent Baselines** in a later session). The table now has **Angle** and **Phase DAC (%)** columns, right after Time (Fig. 2).
2. Select the whole baseline run in one go: click its first row and shift-click its last. Mixed angles are expected; don't pick them out by hand.
3. Read the line above **Set Baseline**. It says what will be set for each angle (rows, sweeps, DAC setting), and in red what needs attention: an angle with no rows, rows at another tune, or a baseline taken at a setting the tune has since moved from.
4. Click **Set Baseline**. The 0° average becomes the main baseline (as in normal mode) and each other angle gets its own. Each angle's baseline is written to `recent_baselines.txt`, so in a later session **Show Recent Baselines** lists one row per angle: select those rows together and **Set Baseline** again.

5. Check the **Baselines** line on the Run tab, under the Quadrature status: every angle should say ok.

The Baseline tab's plots. While Quadrature is ticked, or a selection holds quadrature rows, the two plots become a 2×2 grid (Fig. 2): the left column is absorptive (0°, with $-1 \times 180°$ on top), the right column dispersive (90°, with $-1 \times 270°$). The top row is Selected Baseline. The bottom row is Current Sweep minus Baseline, using the latest sweep at each angle, so with the line off-window it should be flat noise at every angle, and with the line on it shows each angle's signal. On the top row the $-1\times$ curves sit apart by twice the detector's DC offset, as on the Run tab.

What PyNMR does with the selection.

- Rows are averaged per angle (weighted by sweeps), never across angles.
- Within an angle, only rows at the same phase DAC setting as the latest selected row are averaged, so a selection spanning a retune never mixes tunes; the rows left out are listed.
- Each event subtracts its own angle's baseline. The record says which (`quad_base_ref_deg`), at what setting (`quad_base_vout`), and whether that matches the event's tune (`quad_base_match`).
- An angle with no baseline uses the 0° one, which is wrong for a dispersive record; the Run-tab line shows it in red.
- Selecting an ordinary, untagged baseline (for example one taken with Quadrature off) clears the per-angle ones, and every angle uses it.
- After **Auto Tune** or any change to a tune, that angle's baseline is from another setting: the Run-tab line says "taken at another tune". Take a new baseline set.

6.3 Background subtraction for 90° and 270° records

The Analysis tab's subtraction fits a polynomial to the wings and subtracts it. That suits the absorption, which is small in the wings. The dispersion falls off only as $1/(f - f_L)$, so its tails reach the edges of the window, and a wing fit takes part of them for background. In the 2026-10-27 hall data a cubic wing fit ate the dispersion's tails, while a cubic fitted jointly with the Kramers–Kronig transform of the absorption, over the whole sweep, left a dispersion that matched it to the noise.

PyNMR now does the second for every 90° and 270° record:

$$y(f) = \sum_{j=0}^{3} c_j x^j + s\,\mathcal{H}[A](f), \tag{2}$$

where y is the baseline-subtracted sweep, x the frequency scaled to ± 1 across the window, A the absorption of the latest 0° record of the same run (averaged with $-1\times$ the 180° record of the same cycle in cycle4), $\mathcal{H}$ the Hilbert transform, and s a free scale (the gain ratio and the sign of 90° go into it). Only the cubic is subtracted, so Fit Subtracted keeps the whole measured dispersion. The fitted $s\,\mathcal{H}[A]$ is drawn dashed on the Fit Subtracted Imag plot as a check: with a good Imag tune the two lie on top of each other, and $|s|$ is the gain ratio (about 1).

- 0° and 180° records keep the Analysis tab's subtraction.
- The first dispersive record of a run, before any 0° record, uses the wing fit; `quad_sub_method` records which was used.

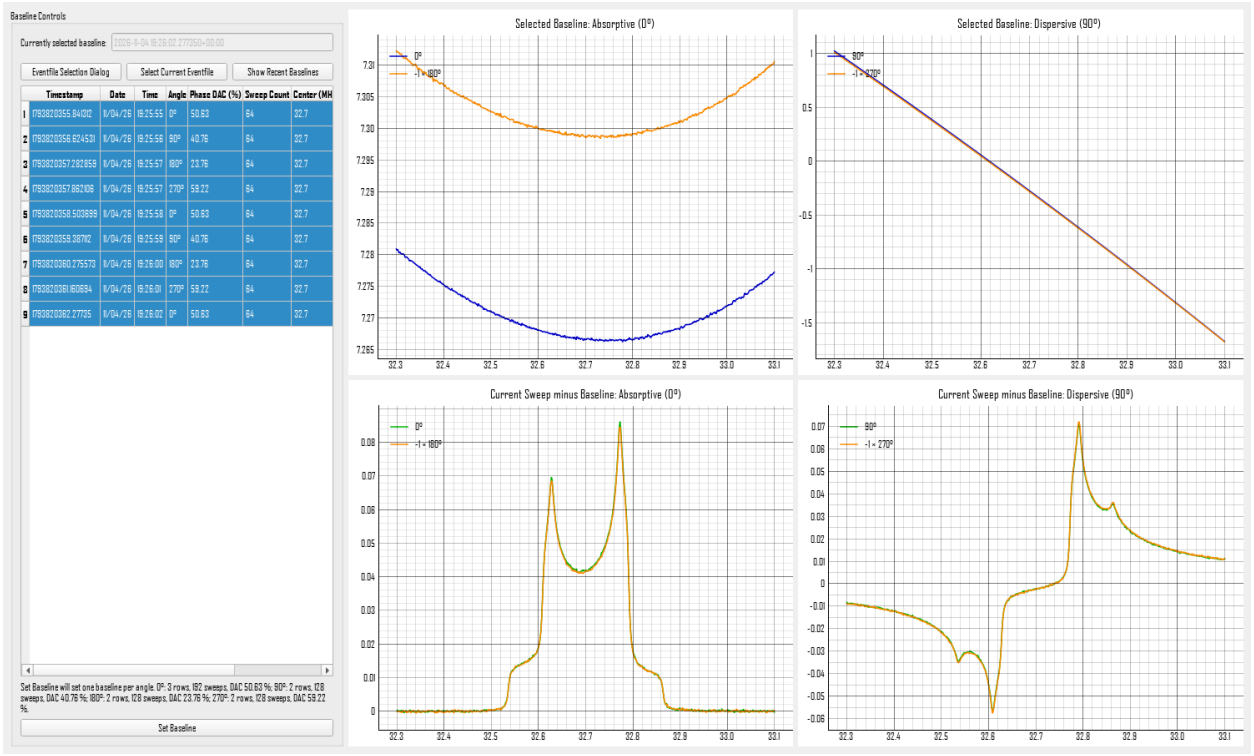


Figure 2: The Baseline tab in the simulator, with a whole cycle4 baseline run (9 rows) selected and set, after the line has come back on. The Angle and Phase DAC columns follow Time. Above **Set Baseline**, the summary lists what each angle gets. Right: the 2×2 grid, absorptive on the left and dispersive on the right; Selected Baseline on top (orange: $-1 \times 180°$ and $-1 \times 270°$), Current Sweep minus Baseline below, where the opposite settings overlay their partners.

- The part of the dispersion due to absorption outside the window can't be predicted from inside it; it is smooth and goes into the cubic.
- `dispersion: joint_kk: False` in `quadrature.yaml` goes back to the wing fit for every angle.

6.4 The plots

While Quadrature is ticked, a 2×3 grid replaces the three plots under the Run tab (Fig. 3).

- The columns are the processing stages: Raw Signal, Baseline Subtracted, Fit Subtracted.
- The top row is Real (0°), the bottom row Imag (90°).
- In cycle4, $-1\times$ the 180° record is drawn in orange over every 0° plot, and $-1\times$ the 270° record over every 90° plot, so the opposite settings can be compared at a glance. On the Raw row they sit apart by twice the detector's DC offset; that gap *is* the offset.
- Baseline Subtracted shows the fitted background (yellow); Fit Subtracted Imag also shows the fitted KK curve, dashed (§6.3).
- A raw curve follows its event live as the sweeps come in. Every other curve changes only when a record at its own angle finishes, so all of them can be watched as the line changes.

- The big polarization number keeps showing the last Real record; the area box notes when the latest record is at another angle.

Unticking Quadrature shows JLab's three plots again.

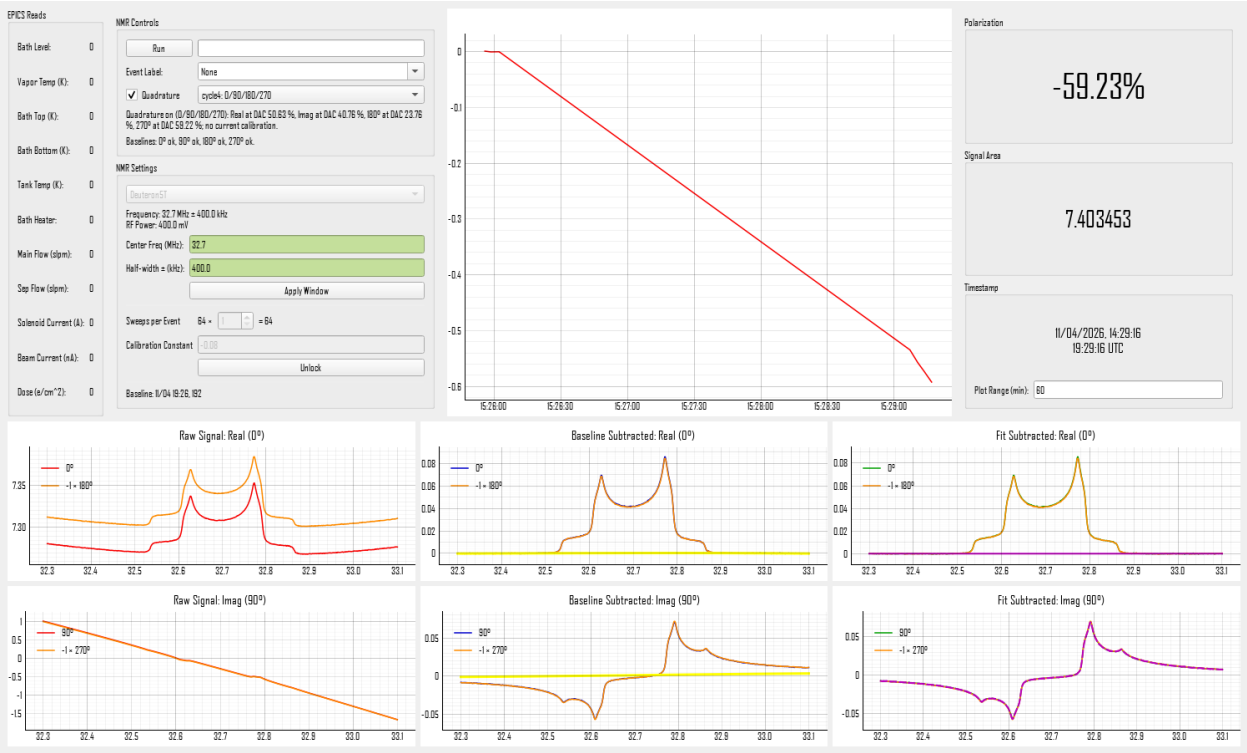


Figure 3: The Run tab during a cycle4 run in the simulator (Sweeps per Event 64×1), after a cycle4 baseline set was selected in one go. Bottom: the 2×3 quadrature grid, with $-1 \times 180°$ and $-1 \times 270°$ in orange. On the Raw row the gap between $0°$ and $-1 \times 180°$ is the simulated detector offset; each angle's own baseline removes it. Fit Subtracted Imag shows the whole dispersion, tails included, with the fitted KK curve dashed on top of it (here $s = 1.01$).

6.5 Where the records go

- **Event file:** every record, at every angle, with the `quad_*` fields of Table 2.
- **EPICS and history (pol plot, TE tab):** $0°$ records only. The area and polarization of $90°$, $180°$ and $270°$ records are not polarizations.
- **Online fitter:** $0°$ records only.

Offline. With cycle4, Eq. (1) gives I and Q free of offsets. With a pair at a known angle δ and gain ratio g (from the calibration), $I = P_0 - o_0$ and $Q = \left[(P_\delta - o_\delta)/g - I\cos\delta\right]/\sin\delta$; the noise penalty is only $1/\sin\delta$, so δ need not be exactly $90°$. The first hall data (2026-10-27) showed the measured dispersion matching the Kramers–Kronig transform of the absorption to about the noise, once the Q-curve background was fitted jointly with it.

Field	Meaning
`quad_ref_deg`	nominal angle: 0, 90, 180 or 270
`quad_vout`	phase DAC value for this record (equal to `phase_vout`)
`quad_tunes`	all tunes in use, e.g. {`"0": 0.514, "90": 0.444, ...`}
`quad_mode`, `quad_sequence`	`pair`/`abba`/`cycle4`/`custom`, and the list of angles
`quad_group`, `quad_index`	one pass through the sequence, and the position in it; pair records by group
`quad_state`	`ok`, or `suspended` if the box refused a DAC write (the rest of the run is Real only)
`quad_dac_ok`, `quad_dac_set`	whether the box accepted the write, and which path set the tune
`quad_settle_ms`	wait after the DAC write
`quad_cal_id`, `quad_cal_valid`	calibration in force, and whether it covers these tunes
`quad_cal_angle_deg`, `quad_cal_gain`	fitted angle and gain of this tune relative to Real
`quad_base_ref_deg`	angle of the baseline subtracted (null for an untagged baseline)
`quad_base_vout`, `quad_base_match`	phase DAC the baseline was taken at, and whether it is this tune
`quad_sub_method`	90°/270° records: `joint KK`, or the wing fit and why
`quad_kk_scale`, `quad_kk_curve`	fitted scale s and $s\,\mathcal{H}[A]$ (dispersive records)

Table 2: Quadrature fields on each event record.

7 Troubleshooting

8 Status and limits

- **Tested on the box (2026-10-27):** quadrature alternation, cycle4, the calibration, the per-angle records. 16 complete cycle4 groups were taken on the deuteron line.
- **Simulator only so far: Auto Tune**, the 2×3 plots, per-angle baseline selection, the joint-KK background for dispersive records, the $64 \times N$ sweep controls, 64-sweep tune chunks, the default tunes, and the fix for 640-sweep timeouts.
- The 90° angle rests on the smooth-gain assumption; 0° and 180° don't.
- The absorptive phase turns across the window by the Q-curve's arc. **Auto Tune** makes the reference absorptive at the centre frequency; the offline complex fit handles the rest.

A Why the Q-curve fixes the absorptive phase

For the constant-current (Liverpool) Q-meter, the detected voltage is a fixed function of the tank impedance, $V = F(Z)$ with $F(Z) = UZ/(R_{cc} + Z)$, and $Z = R + iX(f)$. Sweeping the frequency changes the reactance X; absorption by the sample adds to the coil's resistance R (in proportion to χ''). So

$$\frac{dV}{df} = F'(Z)\, i\, X'(f), \qquad \frac{dV}{d\chi''} \;\propto\; F'(Z)\, \frac{dR}{d\chi''}. \tag{3}$$

Both share the factor $F'(Z)$; one is multiplied by an imaginary number and the other by a real one. The NMR absorption at frequency f therefore points along the *normal* to the Q-curve's trace at f, and the dispersion along the trace itself. The same holds with the $\lambda/2$ cable between tank and coil, to the extent that the cable's own frequency dependence is small over the window.

Symptom	What to do
Quadrature box won't tick	Read the red line beside it: set the named tune on the Tune tab, or move it off the Real tune's DAC value.
Box stops answering ("Exception starting DAQ connection: timed out")	Power cycle the box, then restart PyNMR. It happens when PyNMR dies mid-sweep; check you're on `pynmr312`.
Quadrature suspended mid-run	The box refused a DAC write; the rest of the run is Real only. Check the log, restart the run.
"no current calibration"	The tunes, channel or window changed since the last calibration. The data are fine; run **Auto Calibrate Phase** if you need recorded angles.
Auto Tune not converged	Set the tunes closer by hand (within about 30°) and try again.
Auto Tune result looks wrong	**Undo Auto Tune**. Check the trace plot: a messy trace, or a strong deuteron line, can bias it; try `mask_khz`.
Event of 640 sweeps times out with Quadrature on	Should no longer happen (DAC now written between events); report it with the log.
Baselines line in red	"none": select a baseline set that includes that angle. "taken at another tune": the tune moved (Auto Tune); take a new baseline set.
Imag Fit Subtracted has odd wings	Check `quad_sub_method` is `joint KK` and the dashed KK curve follows the data; a baseline from another tune or a poor 0° record shows up here.
Fitter not updating	Is it running? PyNMR logs one warning per outage and keeps going. Only 0° records are sent.

Table 3: Common problems.

Because F is a Möbius map and X is close to linear in f, the trace is a circle (or, for large R_{cc}, a nearly straight line), and the angle about its centre runs as

$$\theta(f) = \theta_0 + 2\arctan\left(\frac{f - f_0}{\gamma}\right), \tag{4}$$

with f_0 the resonance ($X = 0$) and $\gamma = (R_{cc} + R)/X'$ the width of the trace's parametrization, which is wider than the tank's own half-width R/X'. The normal turns with it, so this is also the absorptive phase at each frequency: the form found experimentally by E. Long years ago. Figure 4 checks this in the simulator's Q-meter with a Q ≈ 26 tank like the Hall B one: the normal and the true absorption direction agree to better than 10^{-6} degrees at every frequency, and both turn by about 19° across a ± 200 kHz window.

Why the centre frequency and not the resonance. A ± 200 kHz window covers a small part of the trace (about $\pm 0.08\,\gamma$ for the Q ≈ 26 circuit, where $\gamma \approx 2.5$ MHz), so f_0 is poorly determined: in a Q ≈ 26 test, the normal taken at the fitted f_0 was 4° off. The normal at the centre frequency needs no f_0 and is where the line sits. Across three test circuits (a straight trace, the Q ≈ 26 circle, and a circle with a $\lambda/2$ cable) and tank detunings up to 200 kHz, it matched the true absorption direction within 0.5°.

B The Auto Tune algorithm

Code: `unh/quadrature/autotune.py` (fits) and `AutoTuneThread` in `unh/quadrature/calibration.py`.

1. **Local calibration.** The steps around the tunes are fitted with the calibration model

$$y_{kj} = g(v_k)\,(1 + d\,\tau_k)\big[\mathrm{Re}\,W_j \cos\phi_k + \mathrm{Im}\,W_j \sin\phi_k\big] + o(v_k),$$

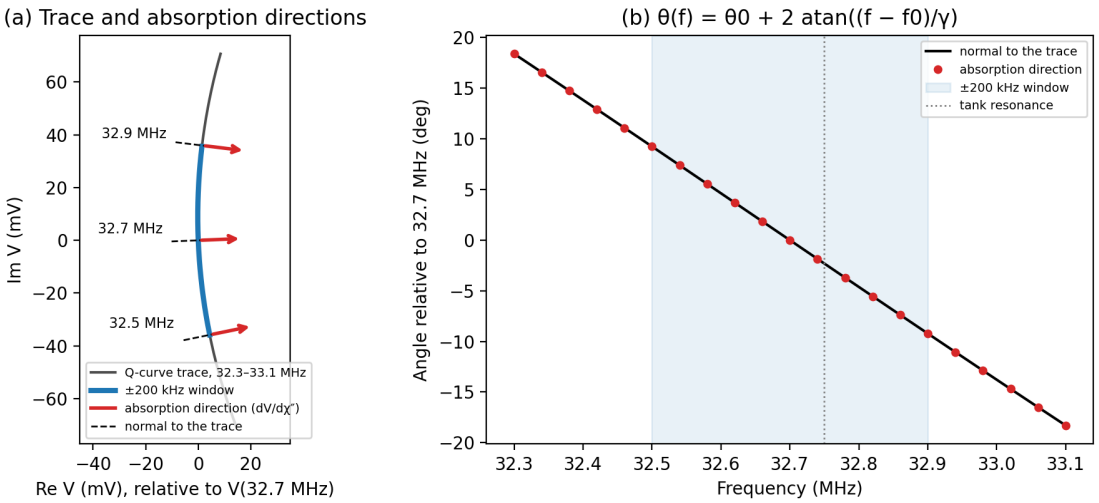


Figure 4: Simulated Q ≈ 26 tank ($R_{cc} = 50\ \Omega$, tank resonance 50 kHz above the window centre). (a) The Q-curve trace with the true absorption direction $dV/d\chi''$ (red arrows) and the normal to the trace (dashes) at three frequencies. (b) Both angles across the sweep, relative to 32.7 MHz.

with the phase ϕ free at every DAC value ($\phi = 0$ at the Real tune), g and o low-order polynomials in v, a linear drift d, and W_j the complex response in 32 frequency bins.

2. **Trace fit.** $W(f)$ is fitted with a circle written to stay finite as it straightens:

$$W(f) = P_0 + i\,e^{i\theta_0}\,s\,\frac{u}{1 - iku}, \qquad u = f - f_0,$$

whose normal at f is at $\theta_0 + 2\arctan\big(k(f - f_0)\big)$, with $k = 1/\gamma$ ($k = 0$ is a line). The normal at the centre frequency, θ_{ref}, and its error come from the fit.

3. **Targets.** 0° is whichever of θ_{ref} and $\theta_{\text{ref}} + 180°$ is nearer the hand Real tune; 90° is whichever of $\theta_{\text{ref}} \pm 90°$ is nearer the hand Imag tune; 180° and 270° are opposite those.

4. **DAC values.** Around each tune, a quadratic $\phi(v)$ through the five settings is solved for the target. If the answer lies outside them, another pass is centred on it (at most `max_passes`, 3).

5. **Polish.** Once every target lies inside its settings, one more pass with `polish_offsets` (±0.5 and ±1 %) is centred on the new tunes, and its answer is final. The spacing of the settings doesn't set the precision (the fit through them does); the polish measures the final tunes directly and shrinks any error from the shape assumed for $\phi(v)$.

Settings are under `autotune:` in `quadrature.yaml`: `offsets`, `sweeps_per_step` (64), `max_passes`, `polish`, `polish_offsets`, `bins`, `gain_order`, `offset_order`, `fit_drift` and `mask_khz`.